

Wave Propagation in Generalized Gyrotropic Media*

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Wave propagation inside generalized gyrotropic media is considered in the present Letter. It is shown that in this type of materials, the left- and right- handed circularly polarized light can be coupled to each other, which may possess some potential applications to quantum information.

Gyrotropic media whose electric permittivity and magnetic permeability are tensors has peculiar optical and electromagnetic properties [1,2]. Recently we proposed an experimentally feasible scheme of testing the so-called quantum vacuum geometric phases of zero-point electromagnetic fields in the helically curved optical fibre by using the gyrotropic-medium fibre [3]. In this Letter, we investigate the wave propagation in a generalized gyrotropic medium with the following permittivity and permeability

$$(\epsilon)_{ik} = \begin{pmatrix} a_e + d_e & b_e - ic_e & 0 \\ b_e + ic_e & a_e - d_e & 0 \\ 0 & 0 & \epsilon_3 \end{pmatrix}, \quad (\mu)_{ik} = \begin{pmatrix} a_h + d_h & b_h - ic_h & 0 \\ b_h + ic_h & a_h - d_h & 0 \\ 0 & 0 & \mu_3 \end{pmatrix}. \quad (1)$$

This type of materials is a generalization of the ordinary gyrotropic medium with [1,2]

$$(\epsilon)_{ik} = \begin{pmatrix} \epsilon_1 & i\epsilon_2 & 0 \\ -i\epsilon_2 & \epsilon_1 & 0 \\ 0 & 0 & \epsilon_3 \end{pmatrix}, \quad (\mu)_{ik} = \begin{pmatrix} \mu_1 & i\mu_2 & 0 \\ -i\mu_2 & \mu_1 & 0 \\ 0 & 0 & \mu_3 \end{pmatrix} \quad (2)$$

and can encompass the electromagnetic parameters of ϵ_{ik} and μ_{ik} in (1) material responses experimentally obtained.

Assume that the wave vector $\mathbf{k}$ of electromagnetic wave propagating inside this medium is parallel to the the third component of the Cartesian coordinate system. According to Maxwellian Equations, one can arrive at

$$-\nabla^2 E_1 = \mu_0 \left(\mu_{21} \frac{\partial^2}{\partial t^2} D_2 - \mu_{22} \frac{\partial^2}{\partial t^2} D_1 \right), \quad -\nabla^2 E_2 = -\mu_0 \left(\mu_{11} \frac{\partial^2}{\partial t^2} D_2 - \mu_{12} \frac{\partial^2}{\partial t^2} D_1 \right), \quad (3)$$

where the electric displacement vector D_1 and D_2 are of the form

$$D_1 = \epsilon_0 (\epsilon_{11} E_1 + \epsilon_{12} E_2), \quad D_2 = \epsilon_0 (\epsilon_{21} E_1 + \epsilon_{22} E_2). \quad (4)$$

Due to the transverse nature of planar electromagnetic wave, the following acquirements are satisfied

$$E_3 = 0, \quad H_3 = 0, \quad \mathbf{k} \cdot \mathbf{E} = 0, \quad \mathbf{k} \cdot \mathbf{H} = 0, \quad \mathbf{k} \cdot \mathbf{D} = 0, \quad \mathbf{k} \cdot \mathbf{B} = 0. \quad (5)$$

Thus with the help of above equations, it is verified that

$$\begin{aligned} \nabla^2 E_1 &= -\epsilon_0 \mu_0 \left[(\mu_{21} \epsilon_{21} - \mu_{22} \epsilon_{11}) \frac{\partial^2}{\partial t^2} E_1 + (\mu_{21} \epsilon_{22} - \mu_{22} \epsilon_{12}) \frac{\partial^2}{\partial t^2} E_2 \right], \\ \nabla^2 E_2 &= \epsilon_0 \mu_0 \left[(\mu_{11} \epsilon_{21} - \mu_{12} \epsilon_{11}) \frac{\partial^2}{\partial t^2} E_1 + (\mu_{11} \epsilon_{22} - \mu_{12} \epsilon_{12}) \frac{\partial^2}{\partial t^2} E_2 \right]. \end{aligned} \quad (6)$$

So,

*It will be submitted nowhere else for the publication, just uploaded at the e-print archives.

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$$\begin{aligned}\nabla^2 \left(\frac{E_1 \pm iE_2}{\sqrt{2}} \right) &= \frac{1}{\sqrt{2}} \epsilon_0 \mu_0 [(\mu_{22}\epsilon_{11} - \mu_{21}\epsilon_{21}) \pm i(\mu_{11}\epsilon_{21} - \mu_{12}\epsilon_{11})] \frac{\partial^2}{\partial t^2} E_1 \\ &+ \frac{1}{\sqrt{2}} \epsilon_0 \mu_0 [(\mu_{22}\epsilon_{12} - \mu_{21}\epsilon_{22}) \pm i(\mu_{11}\epsilon_{22} - \mu_{12}\epsilon_{12})] \frac{\partial^2}{\partial t^2} E_2.\end{aligned}\quad (7)$$

It follows from (1) that

$$\begin{aligned}(\mu_{22}\epsilon_{11} - \mu_{21}\epsilon_{21}) + i(\mu_{11}\epsilon_{21} - \mu_{12}\epsilon_{11}) &= A_+ + iB_+, \\ (\mu_{22}\epsilon_{12} - \mu_{21}\epsilon_{22}) + i(\mu_{11}\epsilon_{22} - \mu_{12}\epsilon_{12}) &= iA_+ + B_+, \end{aligned}\quad (8)$$

where

$$\begin{aligned}A_+ &= a_e a_h - a_e c_h - d_e d_h - i d_e b_h - b_e b_h + i b_e d_h + c_e c_h - c_e a_h, \\ B_+ &= -a_e b_h - i d_e a_h + i d_e c_h - b_e c_h + b_e a_h - c_e b_h + i c_e d_h + i a_e d_h.\end{aligned}\quad (9)$$

In the same fashion, one can arrive at

$$\begin{aligned}(\mu_{22}\epsilon_{11} - \mu_{21}\epsilon_{21}) - i(\mu_{11}\epsilon_{21} - \mu_{12}\epsilon_{11}) &= A_- + iB_-, \\ (\mu_{22}\epsilon_{12} - \mu_{21}\epsilon_{22}) - i(\mu_{11}\epsilon_{22} - \mu_{12}\epsilon_{12}) &= -iA_- - B_-, \end{aligned}\quad (10)$$

where

$$\begin{aligned}A_- &= a_e a_h + a_e c_h - d_e d_h + i d_e b_h - b_e b_h - i b_e d_h + c_e c_h + c_e a_h, \\ B_- &= a_e b_h - i d_e a_h - i d_e c_h - b_e c_h - b_e a_h - c_e b_h - i c_e d_h + i a_e d_h.\end{aligned}\quad (11)$$

Hence, the wave equations of left- and right- handed circularly polarized light¹ is written

$$\begin{aligned}\nabla^2 \left(\frac{E_1 - iE_2}{\sqrt{2}} \right) &= \epsilon_0 \mu_0 A_- \frac{\partial^2}{\partial t^2} \left(\frac{E_1 - iE_2}{\sqrt{2}} \right) + i \epsilon_0 \mu_0 B_- \frac{\partial^2}{\partial t^2} \left(\frac{E_1 + iE_2}{\sqrt{2}} \right), \\ \nabla^2 \left(\frac{E_1 + iE_2}{\sqrt{2}} \right) &= \epsilon_0 \mu_0 A_+ \frac{\partial^2}{\partial t^2} \left(\frac{E_1 + iE_2}{\sqrt{2}} \right) + i \epsilon_0 \mu_0 B_+ \frac{\partial^2}{\partial t^2} \left(\frac{E_1 - iE_2}{\sqrt{2}} \right).\end{aligned}\quad (13)$$

It should be pointed out that the interaction between left- and right-handed polarized light arises in Eq.(13), which is analogous to the Josephson's effect and may therefore be of physical interest.

The coupling of left-handed polarized light to the right-handed one can be characterized by the frequency shift Ω_L and Ω_R , namely, the amplitudes of left- and right- handed circularly polarized light propagating along the $\hat{z}$ -direction are written as follows

$$E_L \sim \exp \left\{ \frac{1}{i} \left[(\omega + \Omega_L) t - \sqrt{A_-} \frac{\omega}{c} z + \phi_L \right] \right\}, \quad E_R \sim \exp \left\{ \frac{1}{i} \left[(\omega + \Omega_R) t - \sqrt{A_+} \frac{\omega}{c} z + \phi_R \right] \right\}. \quad (14)$$

Insertion of Eq.(14) into Eq.(13) yields

$$A_- \omega^2 E_L = A_- (\omega + \Omega_L)^2 E_L + i B_- (\omega + \Omega_R)^2 E_R, \quad A_+ \omega^2 E_R = A_+ (\omega + \Omega_R)^2 E_R + i B_+ (\omega + \Omega_L)^2 E_L, \quad (15)$$

¹For simplicity, without loss of generality, it is assumed that the two mutually perpendicular real unit polarization vectors $\vec{\epsilon}(k, 1)$ and $\vec{\epsilon}(k, 2)$ can be taken to be as follows: $\epsilon_1(k, 1) = \epsilon_2(k, 2) = 1$, $\epsilon_1(k, 2) = \epsilon_2(k, 1) = 0$ and $\epsilon_3(k, 1) = \epsilon_3(k, 2) = 0$. Thus by the aid of the formula $\mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t}$ for the electric field strength, in the second-quantization framework one can arrive at [3]

$$\begin{aligned}E_R &= \frac{E_1 + iE_2}{\sqrt{2}} = i \int d^3 \mathbf{k} \sqrt{\frac{\omega}{2(2\pi)^3}} [a_L(k) \exp(-ik \cdot x) - a_R^\dagger(k) \exp(ik \cdot x)], \\ E_L &= \frac{E_1 - iE_2}{\sqrt{2}} = i \int d^3 \mathbf{k} \sqrt{\frac{\omega}{2(2\pi)^3}} [a_R(k) \exp(-ik \cdot x) - a_L^\dagger(k) \exp(ik \cdot x)].\end{aligned}\quad (12)$$

and consequently

$$A_+ A_- (2\omega\Omega_R + \Omega_R^2) (2\omega\Omega_L + \Omega_L^2) + B_+ B_- (\omega + \Omega_L)^2 (\omega + \Omega_R)^2 = 0, \quad (16)$$

which is a restricted condition regarding the frequency-shift relation between left- and right-handed circularly polarized light.

Note that in the case of conventional gyrotropic media characteristic of such permittivity and permeability tensors (2), where $d_e = b_e = d_h = b_h = 0$, $a_e = \epsilon_1$, $c_e = -\epsilon_2$, $a_h = \mu_1$, $c_h = -\mu_2$, the optical refractive index squared is of the form

$$A_{\pm} = (\epsilon_1 \pm \epsilon_2) (\mu_1 \pm \mu_2), \quad (17)$$

and the coupling coefficients $B_{\pm}$ are vanishing.

More recently, one of the artificial composite metamaterials, the left-handed medium which has a frequency band (GHz) where the electric permittivity (ϵ) and the magnetic permeability (μ) are simultaneously negative, has focused attention of many authors both experimentally and theoretically [4–8]. In the left-handed medium, most phenomena as the Doppler effect, Vavilov-Cherenkov radiation and even Snell's law are inverted. In 1964², Veselago first considered many peculiar optical and electromagnetic properties, phenomena and effects in this medium and referred to such materials as left-handed media, since in this case the propagation vector $\mathbf{k}$, electric field $\mathbf{E}$ and magnetic field $\mathbf{H}$ of light wave propagating inside it form a left-handed system [1]. It follows from Maxwell's curl equations that such media having negative simultaneously negative ϵ and μ exhibit a negative index of refraction, *i.e.*, $n = -\sqrt{\epsilon\mu}$. In experiments, the negative ϵ and μ can be respectively realized by using a network (array) of thin (long) metal wires [9] and a periodic arrangement of split ring resonators [6]. A combination of the two structures yields a left-handed medium. Veselago's original paper and most of the recent theoretical works discussed mainly the characteristics of electromagnetic wave propagation in isotropic left-handed media, but up to now, the left-handed media that have been prepared successfully experimentally are actually anisotropic in nature, and it may be very difficult to prepare an isotropic left-handed medium [10]. Hu and Chui presented a detailed investigation on the characteristics of electromagnetic wave propagation in uniaxially anisotropic left-handed media [10]. It is believed that the above-presented treatment of wave propagation in generalized gyrotropic media may be applicable to the uniaxially anisotropic left-handed materials.

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- [1] Veselago, V. G., *Sov. Phys. Usp.* **10**, 509 (1968).
- [2] Li K. and Pan W. Y., *Chin. Phys.* **11**, 1245 (2003).
- [3] Shen, J. Q., *arXiv: quant-ph/0304172* (2003).
- [4] Smith, D. R., Padilla, W. J., Vier, D. C. *et al.*, *Phys. Rev. Lett.* **84**, 4184 (2000).
- [5] Klimov, V. V., *Opt. Comm.* **211**, 183 (2002).
- [6] Pendry, J. B., Holden, A. J., Robbins, D. J. and Stewart, W. J., *IEEE Trans. Microwave Theory Tech.* **47**, 2075 (1999).
- [7] Shelby, R. A., Smith, D. R. and Schultz, S., *Science* **292**, 77 (2001).
- [8] Ziolkowski, R. W., *Phys. Rev. E* **64**, 056625 (2001).
- [9] Pendry, J. B., Holden, A. J., Robbins, D. J. and Stewart, W. J., *J. Phys.: Condens. Matter* **10**, 4785 (1998).
- [10] Hu, L. B. and Chui, S. T., *Phys. Rev. B* **66**, 085108 (2002).

²Note that, in the literature, several authors mention the year when Veselago suggested the *left-handed media* by mistake. They claim that Veselago proposed the concept of *left-handed media* in 1968. On the contrary, the true fact is as follows: Veselago's excellent paper was first published in July, 1964 [*Usp. Fiz. Nauk* **92**, 517-526 (1964)]. In 1968, this original paper was translated into English by W. H. Furry and published again in the journal of *Sov. Phys. Usp.* [1].